

Scrambled when wrong, left alone when right: a prediction-free account of learning in DishBrain

Konstantin Kladko

Abstract

Cortical cultures improve at Pong when misses are followed by unpredictable stimulation, which has been read as support for the free-energy principle [1]. But the post-miss stimulus is also more disruptive: twice the amplitude and followed by eight seconds without input. We show that this alone suffices: if failure perturbs the network more than success, the network stays in each setting in inverse proportion to its failure rate, and improves with no prediction or reward. A closed-loop model reproduces the reported conditions and predicts anti-learning when success is more disruptive. We propose experiments that separate the two accounts.

1 Two differences, one interpretation

In DishBrain [1], after a hit the culture receives a “predictable” stimulus: 100 Hz for 100 ms on all eight electrodes at the sensory amplitude, 75 mV. After a miss it receives an “unpredictable” one: 150 mV at 5 Hz, at random sites and times, for 4 s, then 4 s without stimulation. The authors chose the higher voltage so that it would force action potentials “regardless of the state the cell was in, thereby being even more disruptive to the culture” [1]. The post-miss feedback is therefore both less predictable and more disruptive, and the published controls do not separate the two: *No feedback* removes both differences, and *Silent* still withdraws the sensory input for eight seconds after a miss only. Earlier criticism concerned the words “intelligence” and “sentience” [2, 3]; our question is which property of the feedback teaches.

2 The scrambler

Let ψ be the network’s setting and $P(\psi)$ its probability of missing. Suppose a miss moves ψ by a symmetric kernel $K(\psi, \psi') = K(\psi', \psi)$ and a hit leaves it in place. Nothing tells the network which way to change.

Proposition 1. *If $P > 0$ and K is irreducible, the stationary distribution is $\rho(\psi) \propto 1/P(\psi)$. The long-run miss rate is the harmonic mean of P , which is below its arithmetic mean (the miss rate of a random setting) unless P is constant.*

Proof. $\rho \propto 1/P$ satisfies detailed balance, $\rho(\psi)P(\psi)K(\psi, \psi') = \rho(\psi')P(\psi')K(\psi', \psi)$, because K is symmetric; the rest is the AM–HM inequality. $\square$

A setting that misses ten times less often is kept ten times longer: good settings are not sought, only left alone. In general both outcomes perturb, with variance σ_m^2 after a miss and σ_h^2 after a hit. For small isotropic steps the setting diffuses with $D(\psi) = \frac{1}{2}[\sigma_m^2 P + \sigma_h^2(1 - P)]$, and the Itô form of the diffusion gives $\rho \propto 1/D$ [4].

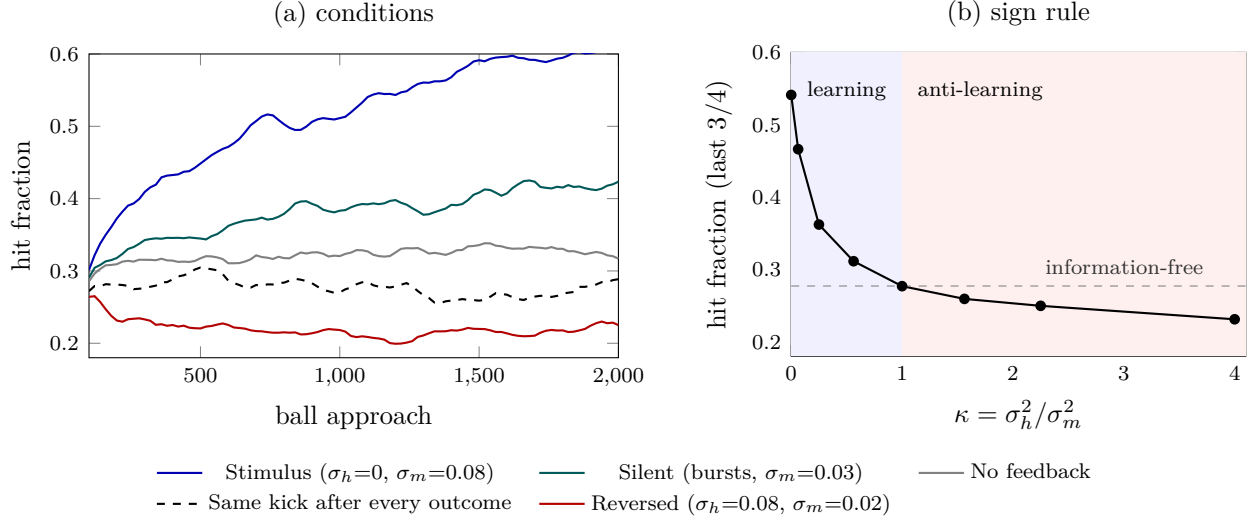


Figure 1: Scrambler learning in a closed-loop Pong model (400 simulated cultures). (a) Hit fraction along a session (sliding window of 100 approaches). (b) Hit fraction over the last three quarters of a session against how much a hit perturbs relative to a miss ($\sigma_m = 0.08$); dashed: the information-free level. Nothing in the model computes a prediction, a reward or the sign of an error.

Proposition 2. With $\kappa = \sigma_h^2/\sigma_m^2$, $\rho \propto 1/[\kappa + (1 - \kappa)P]$. The stationary miss rate is below the uniform average for $\kappa < 1$ (learning), equal to it for $\kappa = 1$, and above it for $\kappa > 1$ (anti-learning).

Predictability does not enter; what matters is how far each outcome moves the network. The principle is old: Ashby’s ultrastability [5], orthokinesis [4], the stimulus regulation principle in cultures [6, 7] and learning by stimulation avoidance [8, 9]. None of these works analyses DishBrain. We add that DishBrain’s own feedback contains this contingency, its quantitative consequences including the sign reversal at $\kappa = 1$, and experiments that decide.

3 A closed-loop model

On a two-parameter paddle model (31×21 grid of settings) Proposition 1 predicts a stationary miss rate of 0.815; simulation gives 0.810, against 0.845 for a random setting. In a DishBrain-like loop (`scrambler.py`) the ball is place- and rate-coded on 8 channels as in the experiment, two motor populations read it through 16 plastic weights, and the only teacher is a random kick to all weights after each approach (s.d. σ_m after a miss, σ_h after a hit). Sessions of 2000 approaches compare the first quarter with the rest, like the first 5 of 20 minutes. Silent is modelled as a weaker scrambler ($\sigma_m = 0.03$ against 0.08), because without stimulation unstimulated cultures burst [10] and bursts make weights drift [11]; this is a testable assumption.

Over 400 simulated cultures the within-session gain in hit fraction is $+0.161 \pm 0.014$ for Stimulus, $+0.068 \pm 0.012$ for Silent and $+0.017 \pm 0.007$ for No feedback: the reported ordering, with Silent above No feedback but weaker [1]. The same kick after every outcome gives -0.010 ± 0.007 , and reversed disruption -0.019 ± 0.006 . Performance falls monotonically with κ and crosses the information-free level at $\kappa = 1$ (Fig. 1b). The model is not fitted to data; the claim is only that the contingency suffices.

Protocol	After a hit	After a miss	Predictability	Scrambler
A. Published	75 mV, 100 Hz, 100 ms	150 mV random, 4 s; 4 s without input	learns	learns
B. Predictable penalty	as A	150 mV, same dose as A, in a fixed sequence	weak or none	learns as A
C. Quenched penalty	as A	8 s of random rapid low-amplitude stimulation that suppresses bursts [10]	learns	weak or none
D. Reversed	150 mV fixed, 1 s	75 mV random, 1 s	learns	below E
E. Information-free	same random stimulus	same random stimulus	none	none

Table 1: Protocols that separate the accounts. B and C are the minimal pair; D tests the sign rule; E is the reference level for D.

4 Experiments that decide

The accounts agree on every condition tested so far and disagree on those in Table 1. B and C each change one property of the post-miss feedback and hold the other; D reverses which outcome is more disruptive.

Two analyses of existing recordings also decide. *Per-event displacement*: the scrambler predicts that a miss moves the network’s state (e.g. the evoked response to a fixed probe) further than a hit, and that across cultures the gain grows with this difference. *Bursts after a miss*: in Silent, the number of spontaneous bursts after misses should predict each culture’s gain.

5 Discussion

If the scrambler account holds, DishBrain shows a generic property of plastic networks under outcome-contingent perturbation rather than predictive processing, though a surprise-minimizing system may implement exactly this. It also explains why gains did not carry over between sessions [1]: a setting held only by the absence of perturbation is diffused by bursting [11]. *Limitations*: stimulation is not isotropic, and a 100 Hz burst may itself potentiate pathways [12]; the displacement analysis measures this directly.

Code availability. `scrambler.py` (standard-library Python) and the data behind Fig. 1 are in `papers/dishbrain-scrambler` of <https://github.com/kladkogex/20-watt-computer>.

References

- [1] B. J. Kagan, A. C. Kitchen, N. T. Tran, F. Habibollahi, M. Khajehnejad, B. J. Parker, A. Bhat, B. Rollo, A. Razi, and K. J. Friston, In vitro neurons learn and exhibit sentience when embodied in a simulated game-world, *Neuron* **110**, 3952–3969 (2022). doi:10.1016/j.neuron.2022.09.001
- [2] F. Balci, S. Ben Hamed, T. Boraud, S. Bouret et al., A response to claims of emergent intelligence and sentience in a dish, *Neuron* **111**, 604–605 (2023).

- [3] B. J. Kagan, A. Razi, A. Bhat, A. C. Kitchen et al., Scientific communication and the semantics of sentence, *Neuron* **111**, 606–607 (2023).
- [4] M. J. Schnitzer, Theory of continuum random walks and application to chemotaxis, *Phys. Rev. E* **48**, 2553–2568 (1993).
- [5] W. R. Ashby, *Design for a Brain* (Chapman & Hall, London, 1952).
- [6] G. Shahaf and S. Marom, Learning in networks of cortical neurons, *J. Neurosci.* **21**, 8782–8788 (2001).
- [7] S. Marom and G. Shahaf, Development, learning and memory in large random networks of cortical neurons: lessons beyond anatomy, *Q. Rev. Biophys.* **35**, 63–87 (2002).
- [8] L. Sinapayen, A. Masumori, and T. Ikegami, Learning by stimulation avoidance: a principle to control spiking neural networks dynamics, *PLoS ONE* **12**, e0170388 (2017).
- [9] A. Masumori, L. Sinapayen, N. Maruyama, T. Mita, D. Bakkum, U. Frey, H. Takahashi, and T. Ikegami, Neural autopoiesis: organizing self-boundaries by stimulus avoidance in biological and artificial neural networks, *Artif. Life* **26**, 130–151 (2020).
- [10] D. A. Wagenaar, R. Madhavan, J. Pine, and S. M. Potter, Controlling bursting in cortical cultures with closed-loop multi-electrode stimulation, *J. Neurosci.* **25**, 680–688 (2005).
- [11] Z. C. Chao, D. J. Bakkum, D. A. Wagenaar, and S. M. Potter, Effects of random external background stimulation on network synaptic stability after tetanization: a modeling study, *Neuroinformatics* **3**, 263–280 (2005).
- [12] Y. Jimbo, T. Tatenno, and H. P. C. Robinson, Simultaneous induction of pathway-specific potentiation and depression in networks of cortical neurons, *Biophys. J.* **76**, 670–678 (1999).